

Genesis of the Jinchuan PGE deposit, China: evidence from fluid inclusions, mineralogy and geochemistry of precious elements

X.-Z. Yang¹, S. Ishihara², and D.-H. Zhao³

¹ Ueyama-sisui Co., Ltd., Sapporo, Japan

² Geological Survey of Japan, Tsukuba, Japan

³ Xi'an Institute of Geology and Minerals Resources, Xi'an, China

Received June 14, 2004; revised version accepted May 24, 2005

Published online October 27, 2005; © Springer-Verlag 2005

Editorial handling: C. Ballhaus

Summary

The Jinchuan deposit is a platinum group element (PGE)-rich sulfide deposit in China. Drilling and surface sampling show that three categories of platinum group element (PGE) mineralization occur; type I formed at magmatic temperatures, type II occurs in hydrothermally altered zones of the intrusion, and type III in sheared dunite and lherzolite. All ore types were analyzed for Os, Ir, Ru, Rh, Pd, Pt and Au, as well as for Cu, Ni, Co and S. Type I ore has $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios of <7 and relatively flat chondrite-normalized noble metal patterns; the platinum group minerals (PGM) are dominated by sperrylite and **moncheite** associated with chalcopyrite, pyrrhotite and pentlandite. Type II has $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios from 40 to 330 and noble metal distribution patterns with a **positive slope**; the most common PGM are sperrylite and Pd **bismuthotelluride** phases concentrated mostly at the margins of base metal sulfides. Type III ores have the highest $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios from 240 to 710; the most abundant PGM are sperrylite and phases of the Pt–Pd–Te–Bi–As–Cl system. It is concluded that the Jinchuan deposit formed as a result of primary magmatic crystallization followed by hydrothermal remobilization, transport, and deposition of the PGE.

Introduction

The Jinchuan intrusion hosts a major source for Ni and Cu in China. It was discovered in 1958 by identification of a surface gossan at the western part of the intrusion. By the early 1970s, most of the chemical and mineralogical characteristics were documented, summarized by *S.G.T.* (1984). Several other studies have

also been published on the general geological background and mineralogy (Si, 1980; Chai and Naldrett, 1994; Yang, 1994, Barnes and Tang, 1999), REE and PGE geochemistry (Sun, 1986; Yang et al., 1991; Chai and Naldrett, 1992a), Rb–Sr and Sm–Nd isotope geology (Tang, 1991). The petrology and geochemistry was discussed by Chai and Naldrett (1992b), Chai et al. (1993), Yang et al. (1994), VanderMeer et al. (1997), Li et al. (2004) and De Wall et al. (2004).

Little information is available on PGM and PGE geochemistry, and there is still debate about the genesis of PGE mineralization. The present authors believe that genetic information can be obtained by reconstructing paragenetic processes, both rock-forming minerals (primary and secondary) and mineralizations of PGE orebodies, particularly from PGM and PGE geochemistry. Fluid-melt inclusions trapped in olivine, pyroxene and tremolite, and microthermometric data are used to delineate the conditions of mineralization.

General geology

The Jinchuan intrusion, Proterozoic in age, is located at the margin of the Sino-Korean massif (Fig. 1). It intruded into late Archean metasedimentary rocks consisting of various gneisses, chlorite-quartz schists, banded marbles and migmatites. The intrusion occurs as a NW oriented lens, over 6,000 m long, approximately 200–300 m wide and more than 1,100 m thick (S.G.T., 1984). Chai and Naldrett (1992b) indicated that the primary magma was a high-MgO basalt.

The intrusion is divided into four segments by E–W to NE–SW trending strike-slip faults. Four mines are located within each of these segments and are referred to as mines III, I, II and IV from west to east (Fig. 1). Most of the intrusion (I and II mines) is exposed, except for the two ends (mines III and IV). Mines I and II host the Ni–Cu sulfide orebodies ⊙ and ⊗. Mines III and IV are separated by faults and form two isolated parts that are covered by Quaternary alluvium.

The intrusion consists of five dominant rock types. In order of decreasing abundance these are, lherzolite, dunite, plagioclase lherzolite, olivine websterite and websterite. These rock types subdivide the intrusion into an upper layer, a lower layer and a marginal layer consisting respectively of ultramafic cumulate (dunite and olivine-rich lherzolite), olivine-poor lherzolite, plagioclase lherzolite and websterite, and websterite and chlorite-tremolite schists. The dominant rock-type in the intrusion is lherzolite.

The primary igneous minerals are chromite followed by olivine, orthopyroxene, clinopyroxene, and finally plagioclase. Chromite occurs mainly as accessory phase, occasionally enched by olivine (Fo_{90-77} ; S.G.T., 1984). Orthopyroxene (En_{88-72}) is less abundant than clinopyroxene ($\text{En}_{45-62}\text{Wo}_{37-45}\text{Fs}_{1-11}$) and augite ($\text{En}_{40.5-43.5}\text{Wo}_{38.5-45}\text{Fs}_{14.5-18.0}$). Plagioclase (average An_{63}) is present as allotypic granular crystals and forms a poikilitic texture with olivine. Secondary minerals include serpentine, magnetite, tremolite, talc, magnesite, calcite, chlorite and clinozoisite. Typical hydrothermal minerals such as calcite and chlorite occur in small branching veinlets or at the terminations of the main veins.

The Jinchuan sulfide orebodies are situated in the middle and lower regions of the intrusion, with major orebodies referred to as ⊙ and ⊗ accounting for more than 90% of the known Ni–Cu reserves in the deposit (Fig. 1). Sulfide minerals generally increase in abundance towards the base of the complex. Six primary and secondary

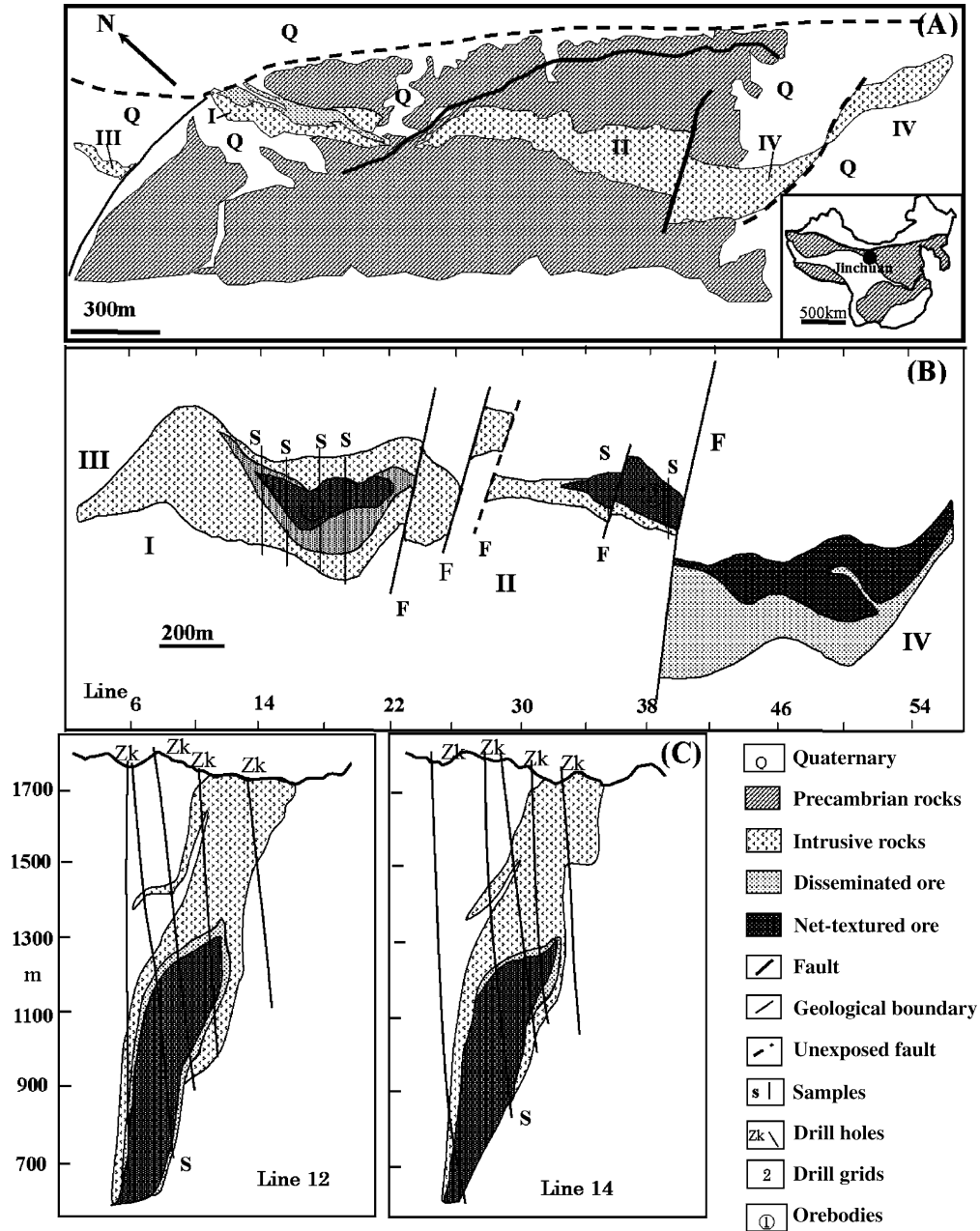


Fig. 1. Location and geology of the Jinchuan deposit (A). Geological map of the 1300 m level (B) and cross sections of drill grids 12 and 14 (C). The drill grid is selectively shown as parallel lines in the upper diagram. The grid lines are at 50 m intervals and are numbered continuously. Zk plus number indicates the drill holes. S indicates sampling locations. I to IV are mines. ⊙ and ⊗ are major ore bodies

textural types of sulfide-PGE ore are recognized, (1) net-textured, (2) disseminated, (3) massive, i.e. >95% sulfide, (4) breccia-type ore, (5) cataclastic and (6) veinlet ores. The dominant Ni sulfide ore is disseminated to net-textured, with sulfide ranging from less than 1 to 40 percent by volume. Massive sulfide ore is rare.

Primary sulfide mineral assemblages consist of pentlandite, chalcopyrite and pyrrhotite. Secondary sulfide minerals include mackinawite + valleriite + pentlandite + chalcopyrite + cubanite + magnetite + violarite, replacing pyrrhotite-pentlandite-chalcopyrite assemblages.

Mode of occurrence of PGE ore bodies

PGE mineralization in the Jinchuan intrusion can be subdivided into three types: I magmatic (Fig. 2A), II hydrothermal (Fig. 2B), and III sheared (Fig. 2C). Type II and III ores formed later than type I ores, based on cross-cutting relationships.

Type I

Type I (magmatic) ore mineralization is sulfide-dominated and occurs within the central part of the intrusion. Sulfide mineralization is typically disseminated to net-textured, with sulfide ranging from 5 to 40 percent by volume (Fig. 2A'). The base-metal mineral assemblage is, in order of decreasing abundance, pentlandite, pyrrhotite and chalcopyrite. The original rock texture is preserved; hydrothermal alteration had very little effect on the sulfide and PGM assemblage. Chromite occurs as subhedral cumulate grains and is fresh.

Platinum-group minerals occur mainly in the magmatic sulfide minerals; only in rare cases they are included in chromite. The proportion of precious metals in the disseminated to net-textured sulfides apparently remained unchanged during metamorphism and serpentinization. Therefore, for Type I deposits, mobility of the metals during serpentinization must have been minor.

Type II

Type II PGE mineralization appears to be concentrated along regional alteration zones, where it is usually associated with base-metal sulfides. The most common form of sulfide is interstitial (up to 35 percent by volume sulfide, Fig. 2B) to serpentine, resulting in a disseminated to net-textured appearance. Original sulfide mineralogy is replaced by mackinawite, millerite, pyrite, cubanite, violarite, bornite, maucherite, nickeline and secondary chalcopyrite (Fig. 2B'), as well as by minerals belonging to the cobaltite-gersdorffite solid-solution series (Mondal et al., 2001; Auge et al., 2002). Pentlandite is commonly rimmed or veined by magnetite. Mackinawite occurs as veinlets or blades within pentlandite. Chalcopyrite occurs as rounded disseminated grains that may contain exsolved cubanite.

Olivine is largely serpentinized (Fig. 2B'). Orthopyroxene is commonly altered to tremolite and lesser amounts of serpentine. Chromite is altered to ferritchromite or/and magnetite.

Type III

Type III ore mineralization is hosted by sheared lherzolite, dunite and brecciated ores and is termed sheared ore (Fig. 2C). The mineralization appears to be concentrated along a regional, complex system of strike-slip faults.

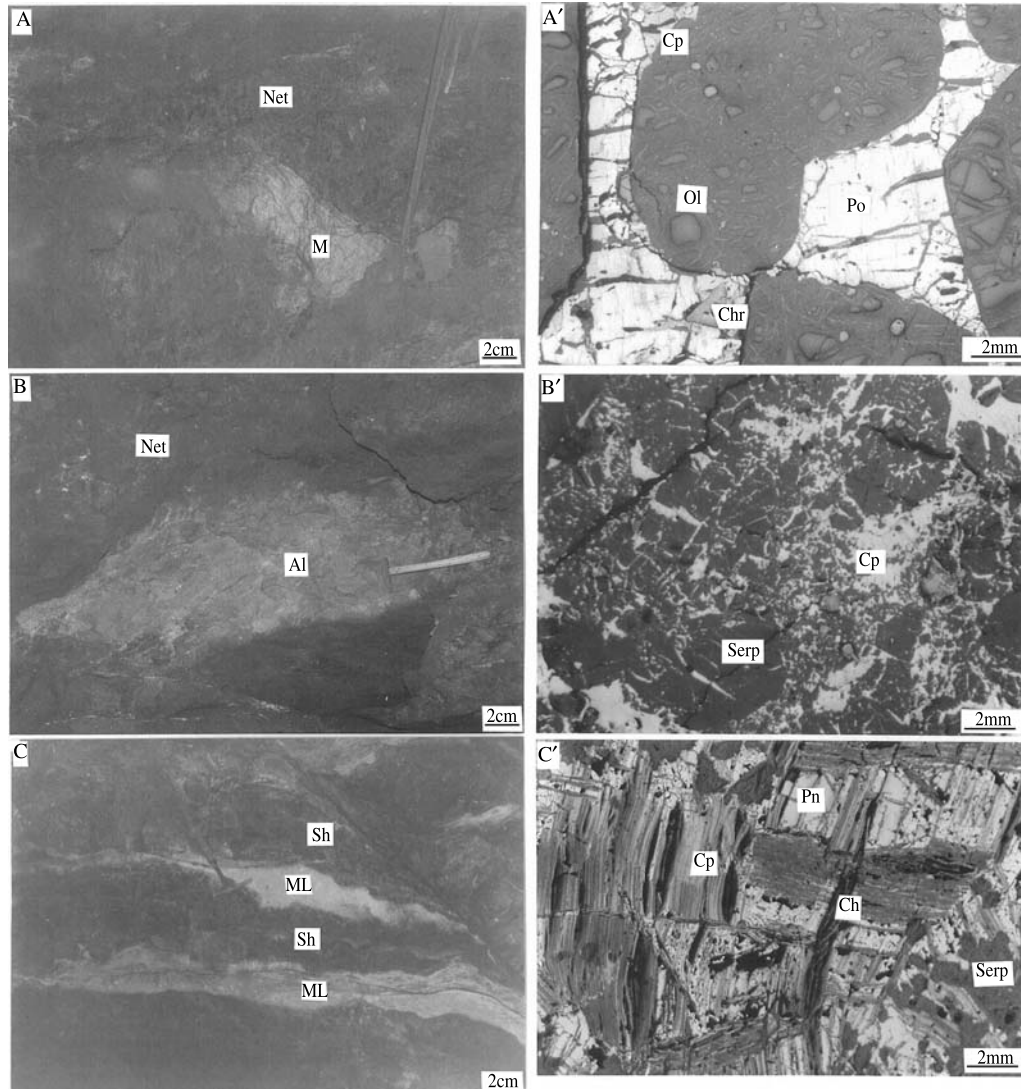


Fig. 2. Modes of occurrence (left) and their photomicrographs (right) of Jinchuan ores. **A** Magmatic ore showing large boulders of net-textured sulfide (Net) cut by a massive sulfide ore (M) dyke (38 line 709 pit, 1300 m level, II mine). **A'** Magmatic ore showing chalcopyrite (Cp) and pyrrhotite (Po) interstitial to olivine (Ol). **B** Hydrothermal ore with hydrous alteration of primary magmatic silicates (1308 level 14 line, II mine). **B'** Hydrothermal ore showing overprint of original ore textures by Cp veins. **C** Detailed view of the sheared ore (Sh) showing a fault, sheared sulfide ore and marble (ML) in sheared lherzolite (1308 level 18 line, II mine). **C'** Sheared ore with cataclastic texture and newly formed chalcopyrite (Cp)

The original texture and mineralogy are replaced by a cataclastic brecciated texture (Fig. 2C'). Silicate minerals in the sheared zone are extensively altered to serpentine, talc, and magnetite. Olivine is completely serpentinized. Most of the chromite is altered to magnetite. The main serpentine mineral around ferrit-chromite or magnetite is antigorite.

Hydrothermal alteration had a strong effect on the sulfide assemblage as suggested by the lack of primary net-texture. Pyrrhotite, pentlandite and chalcopyrite are replaced by mackinawite, pyrite, cubanite, valleriite and violarite. Chalcopyrite occurs as rounded grains, which may contain resolved cubanite. In this case, pentlandite is elongated parallel to the banding, and many fractures, most of which are filled with chlorite, appear perpendicular to the elongation of the crystals. Chalcopyrite occurs predominantly as anhedral assemblages disseminated within other sulfides or as very fine veinlets crosscutting serpentine or tremolite. The sheared zone in the intrusion is greatly enriched in Pt and Pd, with Pt content up to 80 ppm.

Analytical techniques

Analyses of all PGM were carried out by a scanning electron microscope (SEM) equipped with a back-scattered electron detector and by a JEOL-733 electron microprobe (EPMA). Analytical conditions were 15 kV accelerating voltage at 10 nA current on the Faraday cup. Standards used were synthetic PbS for SK_{α} , Au–Ag alloy for AgL_{α} , metallic GaAs for AsL_{α} , and metallic Pt, Pd, Rh, Os, Te, Bi for PtL_{α} , PdL_{α} , RhL_{α} , OsL_{α} , TeL_{α} , BiL_{α} , respectively. CuK_{α} , FeK_{α} , and SK_{α} were calibrated on chalcopyrite.

Concentrations of Ni, Cu and S were determined by X-ray fluorescence at the Xi'an Research Institute of Geology and Mineral Resources, China. PGE and Au were preconcentrated in a sulfide bead following *Robert et al. (1971)*. The concentrations of the noble metals were determined by instrumental neutron activation (NAA) analysis.

A Laser Raman microprobe was used to identify phases in the fluid and melt inclusions. All determinations were conducted at room temperature with a Laser RAMANOR U-1000 instrument. A 400 mW Ar^{+} laser was used to emit a monochromatic beam with a wavelength of 514.5 nm passing through 400 to 500 μm slits. Relative effective cross sections of Raman scattering were used to calculate the mole fraction of each component, according to the formula of *Schrotter and Klockner (1979)*.

Melt inclusions were studied using a Leitz 1350 high-temperature heating stage (*Roedder, 1984*) calibrated on the melting point of Au (1064 °C). During runs, the samples were brought to temperatures between 800 and 1100 °C at 20 °C per minute, and then held at this temperature for 10 to 20 minutes. Inclusions with fluid phases were also studied by low-temperature microthermometry using Chaixmeca and Linkam TH-600 combined heating and cooling stages, calibrated from –190 to 600 °C using various natural and synthetic standards as described by *Roedder (1984)*; pure CO_2 in inclusions in quartz was used for calibration at –56.6 °C.

Platinum-group minerals

Most of the PGM identified are sperrylite and palladium bismuthotelluride phases, which are significant platinum and palladium carriers.

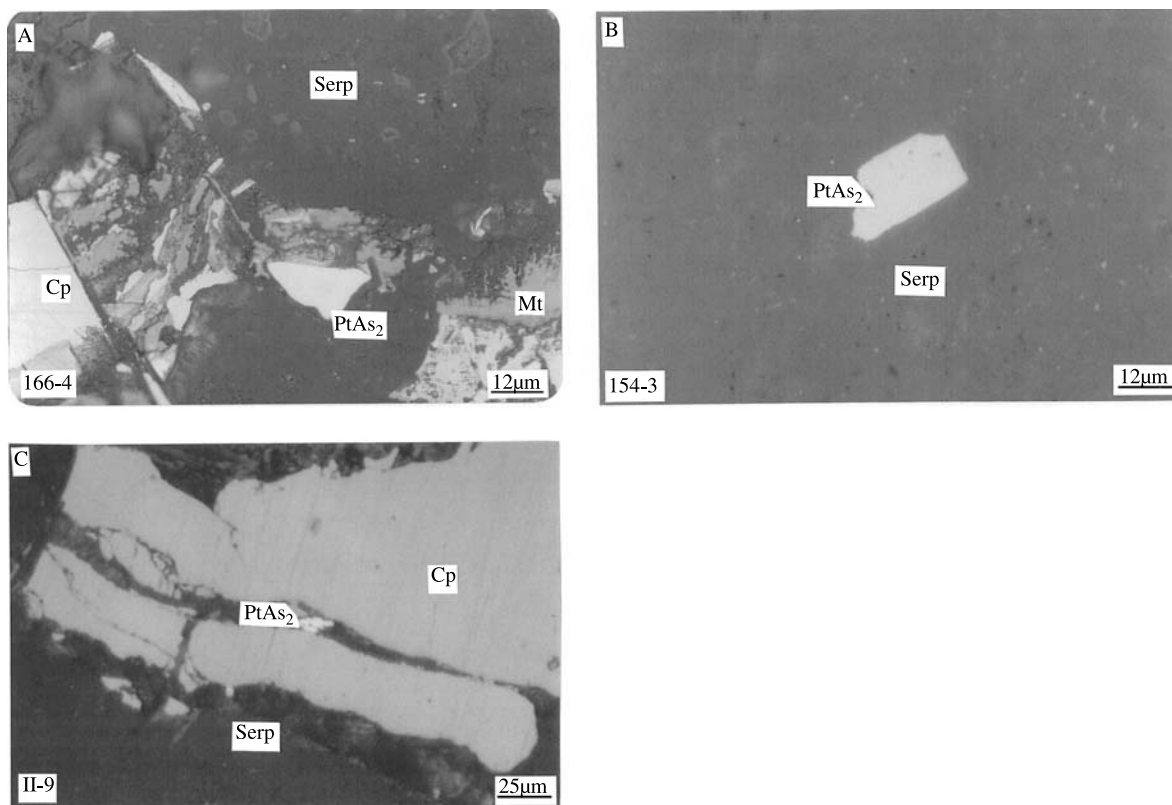


Fig. 3. Examples of sperrylites (PtAs_2) and their host minerals. **A** Anhedral sperrylite with secondary magnetite (Mt) and serpentine (Serp) (Sample 166-4). **B** Euhedral sperrylite in serpentine (Serp) (Sample 154-3). **C** Anhedral sperrylite in fracture of chalcopyrite (Cp) (Sample II-9)

1. Sperrylite

Sperrylite (PtAs_2) is the most common mineral and constitutes about 30 vol% of all PGM. Crystal sizes range from 2 to up to 100 μm in size. Three modes of occurrence are noted; (1) anhedral crystals intergrown with sulfides, magnetite and altered silicate minerals (Fig. 3A); (2) large euhedral crystals occurring in altered silicates (Fig. 3B); and (3) fracture fills within sulfide minerals (Fig. 3C). The largest grains, usually in type II and type III ore, occur preferentially inside serpentine and at the margins of sulfide minerals. In type I ores sperrylite grains are small and mostly included in sulfide. Nine sperrylite grains from different samples were analyzed with EPMA. Compositions are close to stoichiometric PtAs_2 .

2. Moncheite (PtTe_2)

Moncheite is most common in type I to III ores. In type I it is extremely fine-grained (<10 μm ; Fig 4A). Larger grains (4 by 20 μm) were often found in type II

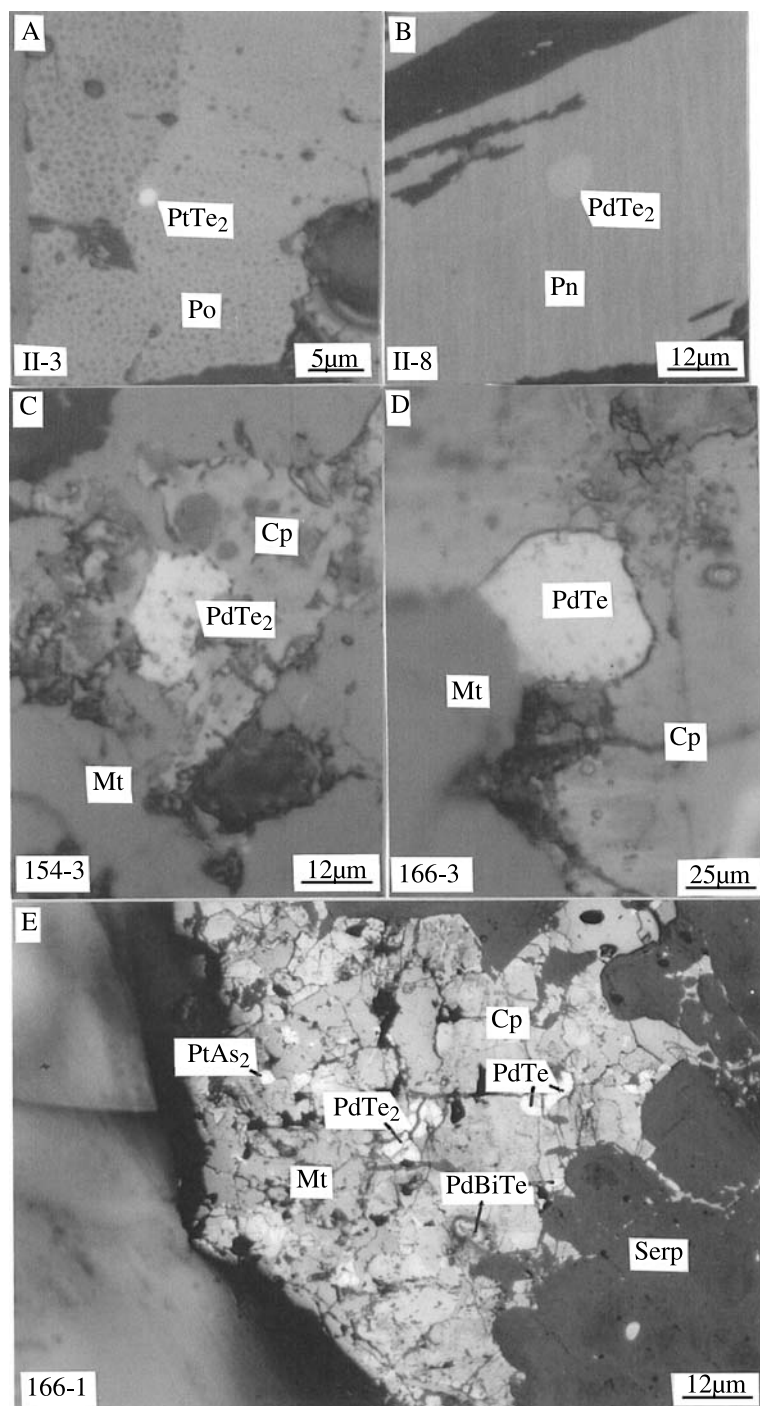


Fig. 4. Photomicrographs of PtTe_2 , PdTe_2 , PdTe and other PGM. **A** Moncheite (PtTe_2) in pyrrhotite (Po) (Sample II-3). **B** Merenskyite (PdTe_2) in pentlandite (Pn) (Sample II-8). **C** Merenskyite (PdTe_2) between secondary magnetite (Mt) and chalcopyrite (Cp) (Sample 154-3). **D** Kotulskite (PdTe) between secondary chalcopyrite (Cp) and magnetite (Mt) (Sample 166-3). **E** Sperrylite (PtAs_2), and merenskyite (PdTe_2) between secondary magnetite (Mt) and chalcopyrite (Cp), kotulskite (PdTe) in a fracture, and michenerite (PdBiTe) at a margin of chalcopyrite (Cp) (Sample 166-1)

Table 1. *Selected results of electron microprobe analyses of moncheite*

Sample	II-7	II-5		II-8		Lb-154		Lb-162		Lb-166	
Anal.	1	2	3	4	5	6	7	8	9	10	11
Pt	42.92	46.21	38.50	38.80	38.56	38.93	36.48	30.82	30.98	25.94	29.87
Pd	–	–	–	–	–	–	–	3.20	2.80	5.20	3.50
Te	56.15	52.94	61.03	44.82	41.51	40.96	41.46	40.68	40.18	53.40	55.10
Bi	0.20	–	–	14.91	19.58	18.85	21.48	24.32	25.79	14.50	10.20
Total	99.27	99.15	99.53	98.53	99.65	98.74	99.42	99.02	99.75	99.04	98.67
Atom. %											
Pt	33.28	36.34	29.21	32.00	32.05	32.67	30.42	25.35	25.47	19.85	22.97
Pd	–	–	–	–	–	–	–	4.83	4.22	7.30	4.93
Te	66.57	63.66	70.79	56.52	52.76	52.56	52.86	51.16	50.51	62.49	64.78
Bi	0.14	–	–	11.48	15.19	14.77	16.72	18.67	19.79	10.36	7.32

– Below detection limit

ore, either included in chalcopyrite, at the contacts of chalcopyrite with hydrous silicate minerals, or in fractures of sulfide minerals. EPMA indicates that moncheite contains small amounts of Bi (Table 1, Fig. 5A).

3. Merenskyite ($PdTe_2$)

Merenskyite forms rounded or subhedral grains 4 to 5 by 35 μm in size and is most common in type II and III ores. Grains are closely intergrown with chalcopyrite, cubanite and/or pentlandite (40 vol%; Fig. 4B and C) or where sulfides are in contact with hydrous silicates (40 vol%), or in fractures within sulfide minerals (20 vol%; Fig. 4E). Merenskyite is mostly intergrown with other PGM. Chemical analyses (Table 2, Fig. 5B) revealed the presence of Ag and Pt, presumably substituting for Pd.

4. Kotulskite ($PdTe$)

Kotulskite occurs as anhedral grains up to 30 μm in size, in both type II and type III ores. Usually it occurs at the margins of chalcopyrite and magnetite grains (20 vol% of the grains; Fig. 4D), within sulfide (20 vol%), or in fractures within sulfide minerals (Fig. 4E, 25 vol%). Microanalysis (Table 2) indicates the presence of Bi and up to 8.8 wt.% Ag.

5. Michenerite ($PdBiTe$)

Michenerite occurs as rounded or subhedral crystals in type II and III ores. In type III ore, it is closely associated with magnetite and serpentine minerals (Figs. 4E and 6). Some grains were found within chalcopyrite and violarite (20 vol% of the grains), but the majority (~80 vol%) is situated at the contacts of sulfide minerals

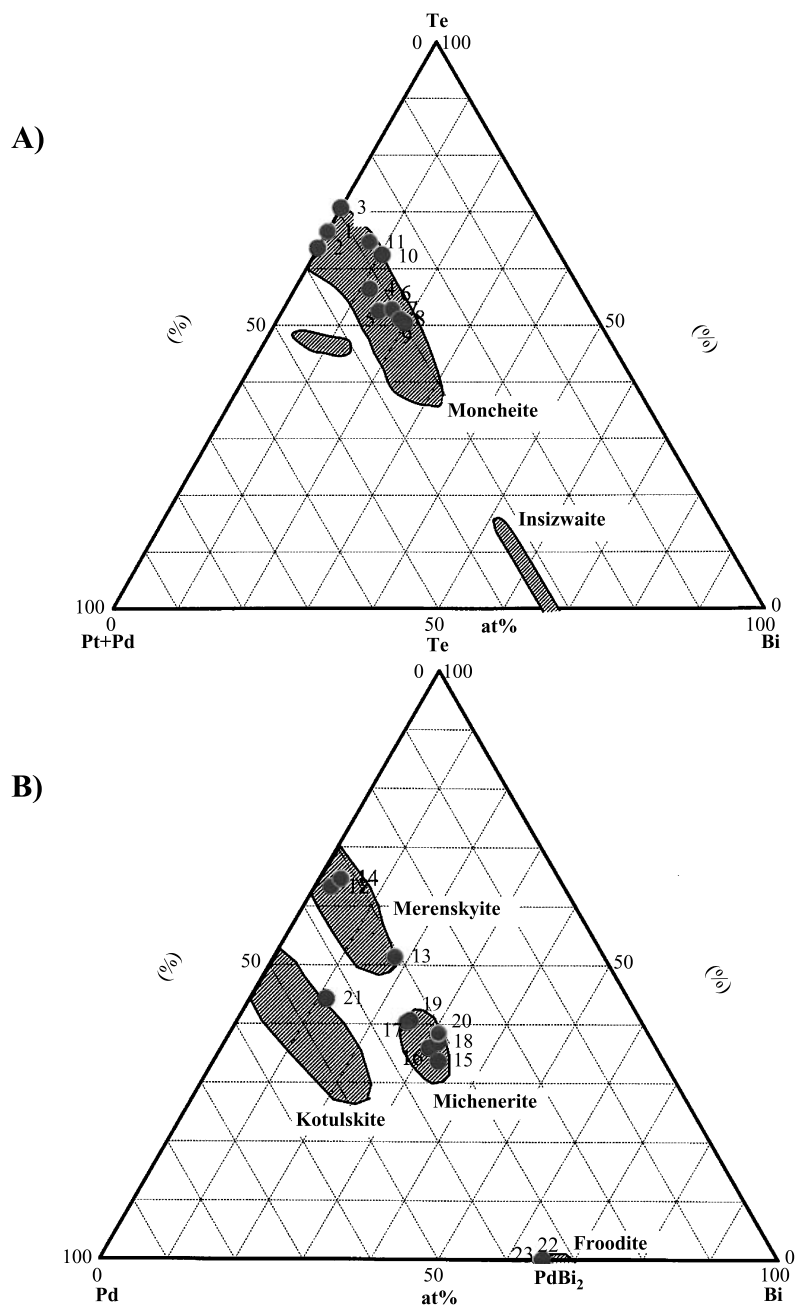


Fig. 5. Chemical compositions of natural Pt–Pd–Te–Bi minerals from the Jinchuan deposit. **A** Pt (+Pd)–Te–Bi diagram showing compositional variation of moncheite. **B** Pd (+Pt, Ag, Fe, Ni, Cu)–Te (+S)–Bi diagram showing compositional variation of merenskyite, kotulskite, michenerite and froodite. Shaded areas represent the compositional fields of Pt–Pd phases; shaded fields: data from *Harney and Merkle (1990)*

with silicate phases. Commonly, michenerite occurs in complex intergrowths with other PGM (froodite etc.). Compositional variability (Table 2) is shown in Fig. 5B. Ag and Pt may substitute for Pd.

Table 2. *Selected electron microprobe analyses of Pd minerals*

Sample	Merenskyite			Michenerite					Kotulskite		Froodite	
	II-7	II-8		II-9			Lb-166		Lb-162		Lb-154	
Anal.	12	13	14	15	16	17	18	19	20	21	22	23
Pt	2.85	10.12	2.95	0.68	2.26	0.15	0.98	1.65	4.47	0.43	1.20	7.30
Pd	21.57	17.43	23.89	18.59	18.81	20.40	21.74	15.36	18.88	27.92	15.80	16.60
Ag	0.89	–	1.59	0.14	0.95	0.00	0.57	8.96	0.33	8.79	4.70	0.40
Fe	1.35	–	–	1.26	0.56	1.30	–	–	–	–	–	–
Ni	0.42	–	–	0.78	0.85	2.35	–	–	–	–	–	–
Cu	1.28	–	–	0.86	0.67	0.00	–	–	–	–	–	–
Te	65.46	45.98	65.39	29.69	31.89	38.00	32.07	35.94	32.48	44.02	0.34	0.00
Bi	4.21	26.11	5.38	47.58	44.46	38.80	44.43	37.01	42.50	18.31	77.93	76.80
S	0.42	–	–	–	–	0.30	–	–	–	–	–	–
Total	98.45	99.64	99.20	99.58	100.45	101.30	99.79	98.92	98.66	99.47	99.97	101.10
Atom. %												
Pt	1.77	7.40	1.91	0.51	1.67	0.10	0.74	1.22	3.46	0.28	1.07	6.63
Pd	24.62	23.37	28.33	25.35	25.44	25.61	30.11	20.78	26.83	33.70	25.88	27.63
Ag	1.00	–	1.86	0.19	1.27	0.00	0.78	11.96	0.46	10.46	7.59	0.66
Fe	2.94	–	–	3.28	1.44	3.11	–	–	–	–	–	–
Ni	0.87	–	–	1.93	2.08	5.35	–	–	–	–	–	–
Cu	2.45	–	–	1.96	1.52	0.00	–	–	–	–	–	–
Te	62.31	51.41	64.66	33.76	35.97	39.78	37.04	40.55	38.49	44.30	0.46	0.00
Bi	2.45	17.82	3.25	33.03	30.61	24.80	31.33	25.49	30.75	11.25	64.99	65.08
S	1.59	–	–	–	–	1.25	–	–	–	–	–	–

– Below detection limit

6. Froodite ($PdBi_2$)

Froodite occurs along fractures (40 vol% of the grains) and at the margins (Fig. 6C) of chalcopyrite, magnetite and pentlandite (60 vol%), mostly in type II and type III ores. Crystals vary in diameter from less than 5 to 35 μm and are usually anhedral in shape. Some Pt substitutes for Pd (Table 2, Fig. 5B) but Te concentrations are lower than in other PGE deposits (cf. Volborth et al., 1986).

7. Ag–Pd–Te–Bi compounds

An Ag-rich phase was detected in type III ore. The mineral occurs in fractures of sulfide and serpentine minerals. The analysis with the highest Ag contents (18 wt.%), also showed 16.4 wt.% Pd, 3.3 wt.% Pt, 2.7 wt.% Ni, 1.3 wt.% Cu, 0.7 wt.% Fe, 46.7 wt.% Te and 11.3 wt.% Bi. The metal to semimetal ratio is close to unity, and a possible formula is $(\text{Ag}_{0.40}\text{Pd}_{0.37}\text{Ni}_{0.11}\text{Cu}_{0.05}\text{Pt}_{0.04}\text{Fe}_{0.03})$ $(\text{Te}_{0.87}\text{Bi}_{0.13})$. It is worth noting that no Ag-rich phases were found in type I ore. Two tiny inclusions (2 to 4 μm) were encountered in serpentine from sample 166 (Fig. 6A) but these grains could not be analyzed quantitatively.

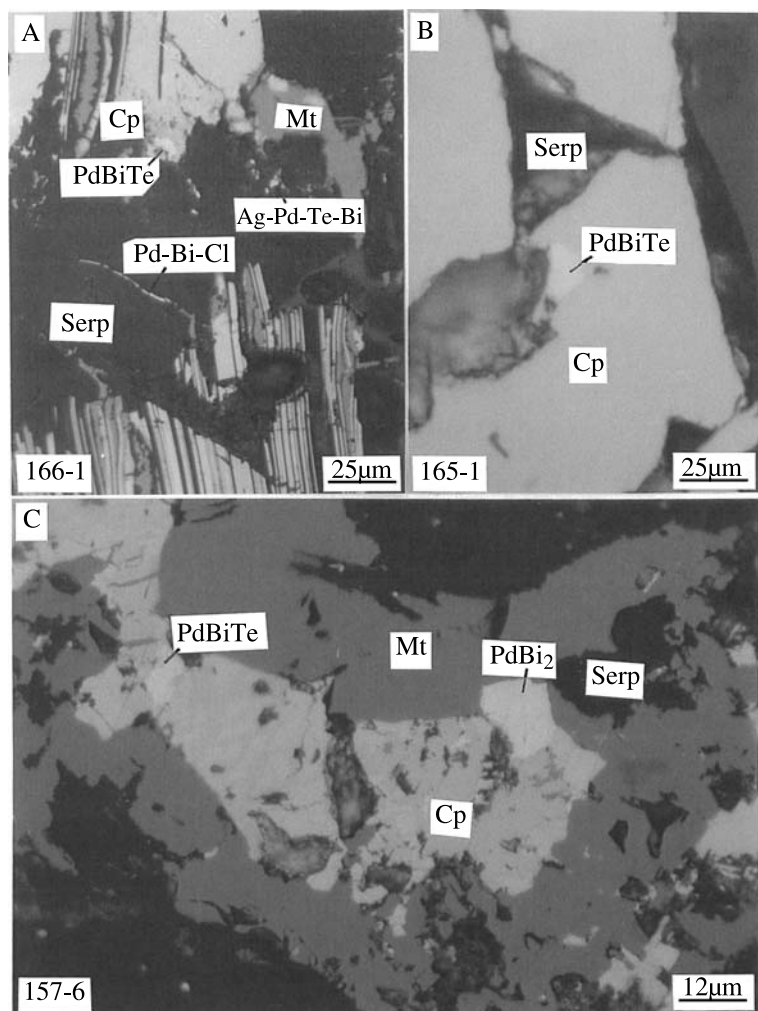


Fig. 6. Photomicrographs of PdBiTe, PdBi₂, and other PGM with their host minerals. **A** Michenerite (PdBiTe) between chalcopyrite (Cp) and serpentine (Serp), Ag–Pd–Te–Bi minerals in serpentine, Pd–Bi–Cl mineral in a fracture through serpentine (Sample 166-1). **B** Michenerite (PdBiTe) in fracture of chalcopyrite (Cp) (Sample 165-1). **C** Froodite (PdBi₂) and michenerite (PdBiTe) between chalcopyrite (Cp) and magnetite (Mt) (Sample 157-6)

Table 3. *Electron microprobe analyses of Pd–Bi–Cl minerals*

Sample	Lb-162		II-7	
	1	2	3	4
Anal.				
Pt	0.99	0.95	1.28	1.22
Pd	27.42	26.84	27.95	28.44
Te	0.35	0.33	0.29	0.21
Bi	59.21	58.93	55.87	55.98
Cl	5.43	4.87	5.64	5.01
Total	92.41	90.97	89.75	89.64

Table 4. Concentrations of precious elements and Ni, Cu, Co and S in Jinchuan ore types

Type ore	Ni	Cu	Co	S	Os	Ir	Ru	Rh	Pt	Pd	Au	Cu/ (Cu + Ni)	Pt/Pd	Pt + Pd	Os + Ir + Ru + Rh	(Pt + Pd)/ (Os + Ir + Ru + Rh)	
Type I Magmatic ore																	
1	II-3	1.66	0.67	0.05	7.05	0.104	0.098	0.100	0.032	0.290	0.160	0.100	0.288	1.813	0.450	0.334	1.347
2	II-5	2.06	0.78	0.05	7.71	0.020	0.024	0.018	0.014	0.170	0.340	0.200	0.275	0.500	0.510	0.076	6.711
3	Lb-452	1.79	0.62	0.03	6.40	0.060	0.070	0.050	0.036	0.020	0.180	0.240	0.257	0.111	0.200	0.216	0.926
4	Lb-426	2.88	0.79	0.07	10.15	0.078	0.090	0.048	0.026	0.020	0.140	0.280	0.215	0.143	0.160	0.242	0.661
Type II Hydrothermal ore																	
5	II-7	2.11	2.25	0.05	7.44	0.002	0.002	0.001	0.001	1.540	0.460	0.300	0.516	3.348	2.000	0.006	333.333
6	II-8	1.59	3.14	0.04	6.79	0.001	0.003	0.001	0.001	0.390	0.470	1.200	0.664	0.830	0.860	0.006	143.333
7	II-13	2.01	1.91	0.06	8.65	0.008	0.007	0.005	0.004	0.660	0.330	0.390	0.487	2.000	0.990	0.024	41.250
Type III Sheared ore																	
8	Lb-154	0.87	3.24	0.02	5.92	0.006	0.008	0.003	0.001	4.020	0.339	1.971	0.788	11.858	4.359	0.018	242.167
9	Lb-162	2.76	1.29	0.05	8.53	0.005	0.006	0.008	0.011	10.630	0.264	0.395	0.319	40.265	10.894	0.030	363.133
10	Lb-166	0.92	3.38	0.02	7.28	0.003	0.003	0.003	0.007	5.735	0.446	0.455	0.786	12.859	6.181	0.016	386.313
11	II-9	1.73	3.13	0.04	7.61	0.001	0.002	0.001	0.001	3.070	0.480	0.400	0.644	6.396	3.550	0.005	710.000

PGE and Au in ppm; Cu, Ni, Co and S in wt. %

8. Unnamed Pd–Bi–Cl minerals

Two irregular grains of an unnamed Pd–Bi–Cl compound are present in Type III ore. One grain occurs in a fracture within serpentine (Fig. 6A). The second grain is intergrown with micheneite and an unnamed Ag–Pd–Te–Bi mineral. In reflected light, this compound is pinkish brown to pale brown. EPMA shows that Pd–Bi–Cl minerals have higher Cl contents (Table 3).

Geochemistry of PGE, Au, Ag, Cu and Ni

Eleven samples from the Jinchuan deposit were analyzed for precious elements, Ni, Cu, Co and S. The samples selected are believed to be representative of the ore types I, II and III, as defined above. Analytical results are listed in Table 4. Most of the analyzed samples show significant enrichments in Pd, Au, and Pt, notably in type II

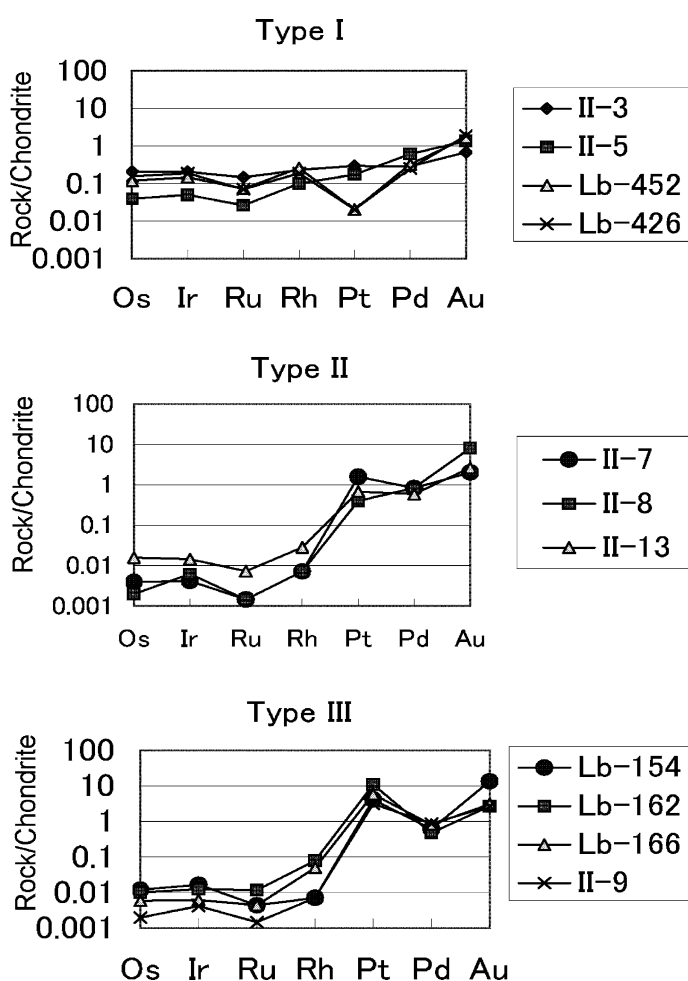


Fig. 7. Chondrite-normalized platinum-group element patterns for magmatic (type I), hydrothermal (type II) and sheared (type III) ores. Chondritic values used for normalization are from *Palme and O'Neill (2004)*

and III ores. Os, Ir, Ru and Rh concentrations are low, sometimes near their respective detection limits. PGE concentrations are normalized to chondritic concentrations (Fig. 7), taken from *Palme and O'Neill (2004)*. The following trends are noted:

Type I ore is characterized by a relatively flat PGE and Au distribution (Fig. 7) and $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh}) < 10$ (Table 4). The PGE and Au curves for individual samples are mostly parallel, indicating similar interelement ratios.

Type II is characterized by noble metal patterns with positive slopes, i.e. uniformly low Os, Ru, Rh and Ir concentrations coupled with enrichments in Pt, Pd and Au (Fig. 7), leading to $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios ranging from 40 to 330. Concentrations in the refractory PGE are low, indicating ores affected by hydrothermal alteration. They are depleted in Os, Ir, Ru, Rh and enriched in Pt, Pd and Au.

Type III ore is also characterized by noble metal patterns with positive slopes (Fig. 7). $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios are highest and range from 240 to 710. Total Os–Ir–Ru–Rh concentrations are near the detection limit.

Fluid-melt inclusions

Four types of fluid-melt inclusions were identified in olivine, pyroxene, apatite, tremolite and hornblende from the Jinchuan deposits. Type A, crystallized polyphase inclusions (+shrinkage bubbles); type B, glass inclusions (+shrinkage bubbles); type C, gas-rich fluid inclusions; type D, aqueous inclusions.

- Type-A polyphase melt inclusions with olivine, orthopyroxene, spinel, unidentified silicate and sulfide as daughter phases, coexist with a fluid phase (Fig. 8A). They homogenize to silicate melt between 950 and 1050 °C, with incipient partial melting noted at around 900 °C. One inclusion rich in sulfide daughter minerals homogenized at >1075 °C.
- Type-B glass inclusions with silicate glass coexisting with sulfide and a fluid phase (Fig. 8B). This inclusion type is found arranged in planes cross-cutting subgrain boundaries in olivine, orthopyroxene and clinopyroxene. The majority homogenized into silicate melt from 650 to 720 °C, with incipient partial melting at 600 °C.
- Type C gas-rich fluid inclusions; compositions include CO₂, CO₂–SO₂–CH₄–N₂–H₂S, and CO₂–H₂O, all with NaCl-undersaturated brine as additional phase (Fig. 8C). Most CO₂ inclusions exhibit melting temperatures at –56.6 °C, ruling out significant concentrations of CH₄, N₂ or other gases. Homogenization into the liquid phase occurred between 22 and 31 °C. CO₂–SO₂–CH₄–N₂–H₂S inclusions exhibit final melting of solid CO₂ between –91.5 and –58.8 °C. Laser Raman analysis indicates the presence of 15 to 16 mol% N₂, 12 to 13 mol% SO₂, 1 to 6 mol% CH₄ and minor H₂S, in addition to CO₂. Homogenization into the liquid phase was noted between 10 and 29 °C. CO₂–H₂O inclusions consist of vapor CO₂ and NaCl-undersaturated brine. Total homogenization temperatures (Th) were not determined as decrepitation occurred above 230 °C.
- Type D aqueous inclusions are two to three phase at room temperature and contain vapor, brine and a daughter mineral (Fig. 8D). Temperature of first melting was noted between –36 and –45 °C, indicating the presence of CaCl₂

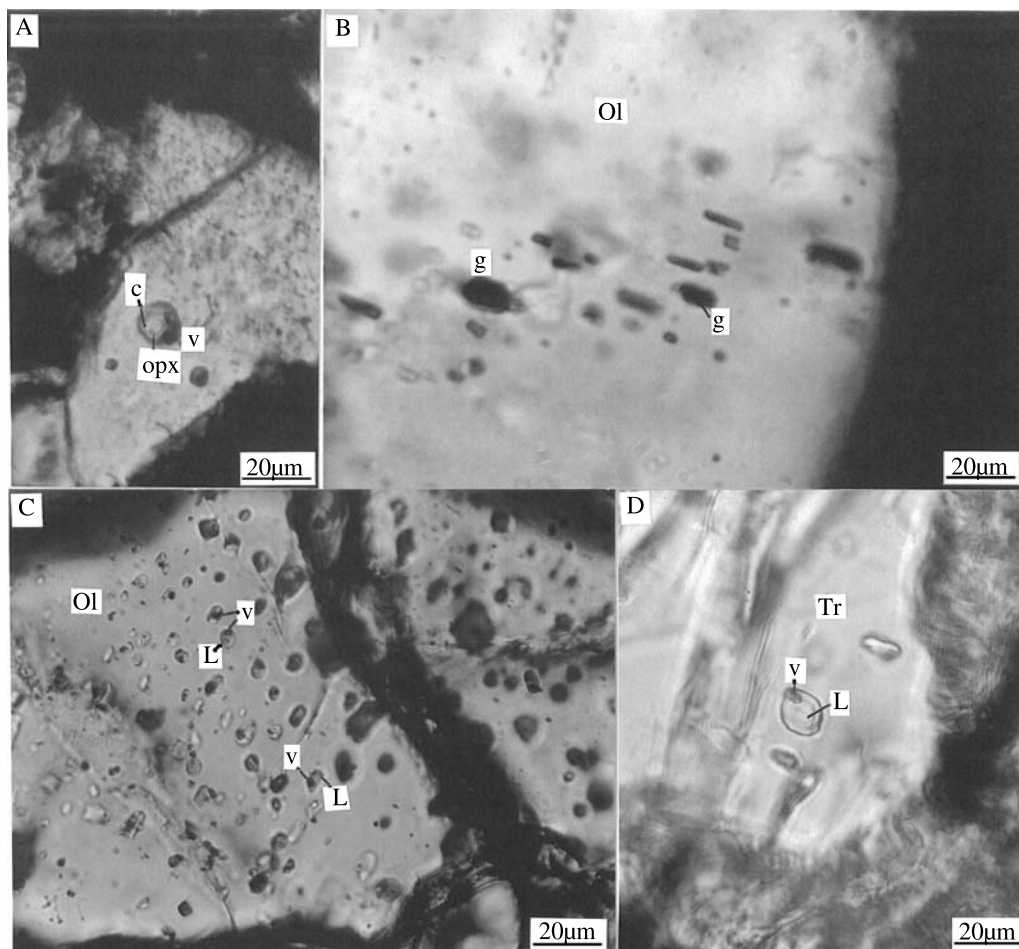


Fig. 8. Photomicrographs of melt and fluid inclusions in silicate minerals. **A** Crystallized polyphase inclusions with opx and an unidentified phase in host olivine; inclusion morphologies follow crystal structure of host olivine. **B** Glass inclusions in host olivine. **C** Primary and secondary fluid inclusions in host olivine; primary inclusions with vapor bubbles that do not reach the surface of the host minerals; secondary inclusions occur along fracture of olivine. **D** Aqueous inclusions in host tremolite

in addition to NaCl (Nyman et al., 1990; Oakes et al., 1990). Type D inclusions are high-salinity inclusions and only observed in hydrothermally overprinted ore. Total homogenization temperatures, determined for few inclusions, range from 235 to 365 °C, but most inclusions decrepitated at above 230 °C. This temperature range is typical for greenschist facies metamorphism.

Discussion and conclusion

PGE–Cu–Ni sulfide deposits are generally considered to be the result of the segregation of sulfide liquid from a mafic or ultramafic magma (Naldrett, 1981; Naldrett and Barnes, 1986). However, hydrothermal PGE deposits have also been

well documented in previous studies (Ballhaus and Stumpfl, 1986; Stumpfl and Ballhaus, 1986; Boudreau et al., 1986; Barkov and Fleet, 2004). Our results suggest three alternative origins in the Jinchuan PGE deposits; a) a magmatic, b) a hydrothermal and c) a tectonic-hydrothermal origin.

A PGM assemblage dominated by sperrylite and moncheite coexisting within base metal sulfides in magmatic cumulate silicate suggests that type I ores crystallized from a sulfide melt. $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios of <7 and flat chondrite-normalized noble metal patterns similar to those of komatiite are considered to indicate that the parent silicate melts originated by extensive degrees of mantle partial melting. The close relationship between Pt and Pd with Ni and Cu demonstrate that the sulfide liquid acted as a collector of PGE derived from the silicate magma. Primary type A inclusions in the olivine indicates that the ultramafic rocks were formed at very high temperatures (above 950°C). Therefore, all of these parameters indicate that type I mineralization is magmatic in origin.

Chai and Naldrett (1992a) maintained that the PGE mineralization of the Jinchuan intrusion was essentially magmatic, based on the close association of noble metal phases with base metal sulfides. However, we note many PGM occurring in serpentine and in fractures of altered minerals, without direct contact with base metal sulfide minerals. Type II ores are thought to be the result of postmagmatic hydrothermal activity. The PGM assemblage, i.e. sperrylite and Pt–Pd–Te–Bi–As phases, is concentrated mostly at the margins of base metal sulfide and they are associated with secondary hydrous alteration phases such as chlorite and serpentine. $(\text{Pt} + \text{Pd})/(\text{Os} + \text{Ir} + \text{Ru} + \text{Rh})$ ratios range from 40 to 330, much higher than in the type I (magmatic) mineralization. Microstructural and mineralogical textures suggest that hydrothermal alteration and mineralization may have been syntectonic. The origin of type B glass inclusions in olivine, with a mean temperature of homogenization of 720 to 650°C , is not clear. Perhaps they record the onset of the hydrothermal overprint. Type II PGE mineralization was probably facilitated by chemical reaction among mineralizing fluids. Oxidation of chromite to magnetite, observed in type II ores, provides evidence for the influx of an oxidizing fluid during PGE deposition. The interaction of oxidized transporting fluids with a reducing environment could produce a decrease in the PGE solubility and cause subsequent precipitation of the ore (Nyman et al., 1990).

Ballhaus and Ulmer (1995) also indicated that almost all of the PGM observed in nature are very late stage, with the exception of some alloys. These are also described in the Baula Complex, India (Auge et al., 2002), and in the Las Aguilas Complex, San Luis Province (Mogessie et al., 2000).

Deformation and brecciation indicate that type III ores experienced some strain. The ores occur as veinlets and in stringers, indicating that PGE mobility may have been enhanced by tectonic as well as hydrothermal fluids. Faulting played a major role in determining the geological structure of the deposit and the location of these types of ores (Yakubchuk and Nikishin, 2004). Fluid mixing coupled with a decrease in temperature and decompression are factors favourable for lowering Pt and Pd solubility and precipitating PGM. The occurrence of a Pd–Bi chloride phase associated with other PGM in the Jinchuan deposit suggests that chloride was an important ligand for PGE. Type C inclusions revealed that the fluid phase included chlorine as well as CO_2 , N_2 , CH_4 and H_2S .

Type D inclusions homogenize from 235 to 365 °C and provide the minimum estimate for the temperature of type III mineralization. This is approximately the same temperature range as for the PGE deposits of the Kuperschiefer in Eastern Europe (*Kucha*, 1982) and PGE mineralization at Coronation Hill, Northern Territory, Australia (*Needham* and *Stuart-Smith*, 1987) but significantly lower than the temperatures of the type II mineralization. These values are in excellent agreement with the minimum temperature of 334 °C reported by *McCallum* et al. (1976) for chalcopyrite exsolution from pyrrhotite.

In summary, fluid phases, evident by hydrothermal alteration and by fluid inclusions, not only altered the primary magmatic minerals but also facilitated sulfide recrystallization. During hydrothermal overprint, PGE initially in solid solution in higher temperature sulfides were mobilized by fluid phases and reprecipitated along the peripheries of sulfide minerals and in fractures of sulfide, in close association with hydrous alteration silicates. The type B and type D fluid and melt inclusions illustrate that hydrous alteration and PGE mobilization took place over a wide temperature interval from ~720 to 235 °C, in close agreement with the hydrothermal Rathbun Lake PGE deposits described by *Rowell* and *Edgar* (1986).

Acknowledgments

We express our sincere appreciation to Prof. *D. X. Yang* for his generous assistance in the study of the Jinchuan deposit. This manuscript was greatly improved by reviews of Dr. *C. Ballhaus* and Dr. *S. Arai*. Associate Editor *C. Ballhaus* is especially thanked for his editorial input and improving the English.

References

- Auge T, Salpeteur I, Bailly L, Mukherjee MM, Patra RN* (2002) Magmatic and hydrothermal platinum-group minerals and base-metal sulfides in the Baula Complex, India. *Can Mineral* 40: 277–309
- Ballhaus CG, Stumpfl EF* (1986) Sulfide and platinum mineralization in the Merensky Reef: evidence from hydrous silicates and fluid inclusions. *Contrib Mineral Petrol* 94: 193–204
- Ballhaus CG, Ulmer P* (1995) Platinum-group elements in the Merensky reef. II. Experimental solubilities of platinum and palladium in Fe_{1-x}S from 950 to 450 °C under controlled $f\text{S}_2$ and $f\text{H}_2$. *Geochim Cosmochim Acta* 59: 4881–4888
- Barkov AY, Fleet ME* (2004) An unusual association of hydrothermal platinum-group minerals from the Imandra layered complex, Kola Peninsula, northwestern Russia. *Can Mineral* 42: 455–467
- Barnes SJ, Tang LH* (1999) Chrome spinels from the Jinchuan Ni–Cu sulfide deposit, Gansu province, People's Republic of China. *Econ Geol* 94: 343–356
- Boudreau AE, Mathez EA, McCallum IS* (1986) Halogen geochemistry of the Stillwater and Bushveld Complexes: evidence for transport of the platinum-group elements by Cl-rich fluids. *J Petrol* 27: 967–986
- Chai G, Naldrett AJ* (1992a) Characteristics of Ni–Cu–PGE mineralization and genesis of the Jinchuan deposit, Northwest China. *Econ Geol* 87: 1475–1495
- Chai G, Naldrett AJ* (1992b) The Jinchuan ultramafic intrusion – cumulate of a high-Mg basaltic magma. *J Petrol* 33: 277–303

- Chai G, Naldrett AJ, Rucklidge JC, Kilian LR (1993) In situ quantitative analyses of PGE and Au in sulfide minerals of the Jinchuan Ni–Cu deposit by accelerator mass spectrometry. *Can Mineral* 31: 19–30
- Chai G, Naldrett AJ (1994) Pyroxene mineral chemistry of the Jinchuan intrusion, China. *Mineral Petrol* 51: 1–20
- De Wall SA, Xu ZH, Li CS, Mouri H (2004) Emplacement of viscous mushes in the Jinchuan ultramafic intrusion, Western China. *Can Mineral* 42: 371–392
- Harney DMW, Merkle RKW (1990) Pt–Pd minerals from the Upper Zone of the eastern Bushveld Complex, South Africa. *Can Mineral* 28: 619–628
- Kucha H (1982) Platinum-group metals in the Zechstein copper deposits, Poland. *Econ Geol* 77: 1578–1591
- Li CS, Xu ZH, De Waal SA, Ripley EM, Maier WD (2004) Compositional variations of olivine from the Ni–Cu sulfide deposit, Western China: implications for ore genesis. *Mineral Deposita* 39: 159–172
- McCallum ME, Loucks RR, Carlson RR, Cooley EF, Doerge TA (1976) Platinum metals associated with hydrothermal copper ores of the New Rambler mine, Medicine Bow Mountains, Wyoming. *Econ Geol* 71: 1429–1450
- Mogessie A, Hauzenberger Ch A, Hoinkes G, Felfernig A, Stumpfl EF, Bjerg EA, Kostadinoff J (2000) Genesis of platinum-group minerals in the Las Aguilas mafic-ultramafic rocks, San Luis Province, Argentina: textural, chemical and mineralogical evidence. *Mineral Petrol* 68: 85–114
- Mondal SK, Baidya TK, Rao KNG, Glascock MD (2001) PGE and Ag mineralization in a breccia zone of the Precambrian Nuasahi ultramafic-mafic complex, Orissa, India. *Can Mineral* 39: 979–996
- Naldrett AJ (1981) Nickel sulfide deposits: classification, composition and genesis. *Econ Geol* 75th Anniv: 628–685
- Naldrett AJ, Barnes SJ (1986) The behavior of platinum group elements during fractional crystallization and partial melting with special reference to the composition of magmatic sulfide ores. *Fortschr Mineral* 64: 113–133
- Needham RS, Stuart-Smith PG (1987) Coronation Hill U–Au mine, South Alligator Valley, Northern Territory: an epigenetic sandstone-type deposit hosted by debris-flow conglomerate. *BMR J Aust Geol Geophys* 10: 121–131
- Nyman MW, Sheets RW, Bodnar RJ (1990) Fluid inclusion evidence for the physical and chemical conditions associated with intermediate temperature PGE mineralization at the New Rambler deposit, southeastern Wyoming. *Can Mineral* 28: 629–638
- Oakes CS, Bodnar RJ, Simonson JM (1990) The system NaCl–CaCl₂–H₂O. I. The ice liquidus at 1 atm total pressure. *Geochim Cosmochim Acta* 54: 603–610
- Palme H, O'Neill HSC (2004) Cosmochemical estimates of mantle composition. In: Carlson RW (ed) *Treatise of geochemistry V21. The mantle and core*. Elsevier, Amsterdam, pp 1–38
- Roedder E (1984) Fluid inclusions. *Mineral Soc Am Short Course* 12: 1–646
- Robert RVD, Van Wyn E, Palmer R (1971) Concentration of the noble metals by a fire-assay technique using nickel sulfide as a collector: South Africa. *Natl Inst Metallurgy Rep* 1371: 1–14
- Rowell WF, Edgar AD (1986) Platinum-group element mineralization in a hydrothermal Cu–Ni sulfide occurrence, Rathbun Lake, northeastern Ontario. *Econ Geol* 81: 1272–1277
- Schrotter HN, Klockner HW (1979) Raman scattering cross sections in gases and liquids. In: Weber A (ed) *Raman spectroscopy of gases and liquids*, chapter 3. Springer, Berlin Heidelberg New York, pp 123–166

- S.G.T. (The Sixth Geological Team of Gansu Geological Survey China)* (1984) Geology of Cu–Ni sulfide deposit in Baijiazuezhi. Geological Publication Bureau, China, pp 225 (in Chinese)
- Si Zhan-Yi* (1980) Characteristics of petrology and mineralogy and their geological senses of some major ore-bearing ultramafic intrusions in Northwest China. *Bull Xi'an Inst Geol Min Res, Chin Acad Geol Sci* 1: 78–89 (in Chinese with English abstract)
- Stumpfl EF, Ballhaus CG* (1986) Stratiform platinum deposits: new data and concepts. *Fortschr Mineral* 64: 205–214
- Sun SX* (1986) The discovery of Pt enrichment parts and existing status of Pt and Pd in mining area of Jinchuan sulfide Ni–Cu deposit. *Geology and Prospecting* 12: 36–39 (in Chinese)
- Tang ZL* (1991) The metallogenetic model of the Jinchuan platinum-bearing copper-nickel sulfide deposit. *International Symposium on Sulfide Deposits, China*, pp 17–38
- VanderMeer F, Fan LH, Bodechtel J* (1997) MAIS imaging spectrometer data analysis for Ni–Cu prospecting in ultramafic rocks of the Jinchuan group, China. *Int J Remote Sensing* 18: 2743–2761
- Volborth A, Tarkin M, Stumpfl EF, Housley RM* (1986) A survey of the Pd–Pt mineralization along the 35 km strike of the J-M Reef, Stillwater Complex, Montana. *Can Mineral* 24: 329–346
- Yakubchuk A, Nikishin A* (2004) Noril'sk-Talnake Cu–Ni–PGE deposits: a revised tectonic model. *Mineral Deposita* 39: 125–142
- Yang XZ* (1994) Genesis of PGE–Cu–Ni ore deposits in the Jinchuan ultramafic intrusion, Gansu Province, China. Thesis, Hokkaido University, 201 p
- Yang XZ, An SY, Hou SJ, Gong ZC, Dong XY* (1991) The REE characteristic and petrogenesis and metallogenetic significance of the Jinchuan ore-bearing ultrabasic intrusion, Gansu Province, China. *J Xi'an College of Geology* 1: 15–22 (in Chinese with English abstract)
- Yang XZ, Matsueda H, Ishihara S* (1994) Mode of occurrence, chemical composition, and origin of Cr–Fe–Ti oxides of the Jinchuan Ni–Cu–PGE deposits, China. *Int Geol Rev* 36: 311–327

Authors' addresses: X.-Z. Yang (corresponding author; e-mail: yang@ueyama-shisui.co.jp), Ueyama-sisui Co., Ltd., N-2 E-13, Sapporo, 060-0032, Japan; S. Ishihara, Geological Survey of Japan, Tsukuba, 305-8567, Japan; D.-H. Zhao, Xi'an Institute of Geology and Minerals Resources, Xi'an, 710054, China